

ORIGINAL ARTICLE

Heart Failure Guideline-Directed Medical Therapy Scoring Systems: A Scoping Review

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BACKGROUND: Guideline-directed medical therapy (GDMT) is central to the care of heart failure with reduced ejection fraction, yet no standard metric exists to quantify its implementation.

METHODS: We conducted a scoping review to catalogue and characterize all published heart failure GDMT scores and summarize their application in clinical studies. We searched MEDLINE, Embase, CENTRAL, and Web of Science from October 2020 to March 2025.

RESULTS: From 544 records, we included 26 studies (19 cohorts, 7 randomized trials; 354 281 patients). Of the 26 studies, 25 (96%) utilized different scores, including 13 adaptations of the Optimal Medical Therapy score originally proposed by the Heart Failure Collaboratory in 2020. All counted RASi (renin-angiotensin system inhibitors), beta-blockers, and mineralocorticoid receptor antagonists; 18 separated sacubitril-valsartan from other RASi, 15 incorporated sodium-glucose cotransporter-2 inhibitors, and 8 captured additional drug classes. Only 2 scores were adjusted for contraindications or intolerance. Across studies, GDMT scores served as a descriptor (N=14), covariate (N=11), predictor of future outcomes (N=12), and end point for GDMT optimization interventions (N=8).

CONCLUSIONS: Multiple, disparate scores have been developed to quantify heart failure with reduced ejection fraction GDMT optimization, undermining the original intent. Key gaps in scores include heterogeneous weighting of drug classes and doses, minimal incorporation of intolerance to identify maximum-tolerated therapy, and heterogeneous incorporation of therapies beyond quadruple therapy. Future studies should focus on clear reporting and justification of the selected GDMT score, and the use of existing scores that incorporate relevant contemporary agents.

Key Words: glucose ■ heart failure ■ hydralazine ■ neprilysin ■ sodium

Guideline-directed medical therapy (GDMT) is the cornerstone of management for heart failure (HF) with reduced ejection fraction (HFrEF).^{1–5} In 2020, the HF collaboratory (HFC) proposed the Optimal Medical Therapy score, later called GDMT score, with the purpose of quantifying background GDMT utilization (ARNI [angiotensin receptor neprilysin inhibitor], ACE inhibitors [angiotensin-converting enzyme], or ARBs [angiotensin receptor blockers], BB [beta-blockers], MRA [mineralocorticoid receptor antagonists], and hydralazine/nitrates) intended to allow comparisons between trial groups and across clinical trials and population-based registries.⁶ In 2022, the score was revised to incorporate additional

pharmacological therapies beyond those included in the original scoring system.⁷

Since the introduction of the original HFC GDMT in 2020, several investigators have proposed variations that expand the HFC GDMT score to incorporate SGLT2is (sodium-glucose cotransporter-2 inhibitors) and other pharmacotherapeutic classes and differ in how they account for contraindications and intolerances, among other changes. This has led to a proliferation of similar but distinct GDMT scores, undermining the original intention of enabling cross-study comparisons. These various GDMT scores have been used for various purposes, including characterizing GDMT utilization patterns,

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WHAT IS NEW?

- In this scoping review of 26 studies, we identified 25 different scores quantifying guideline-directed medical therapy for heart failure with reduced ejection fraction, including 13 adaptations of the heart failure collaboratory score proposed in 2020.
- Available guideline-directed medical therapy scores vary in which medications they include, how they weigh different medications and doses, and how they account for contraindications and intolerances.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Simple guideline-directed medical therapy scores can be useful to describe guideline-directed medical therapy utilization and identify opportunities for optimization.

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
ARB	angiotensin receptor blockers
ARNI	angiotensin receptor neprilysin inhibitor
BB	beta-blockers
GDMT	guideline-directed medical therapy
HF	heart failure
HFC	heart failure collaboratory
HFrEF	heart failure with reduced ejection fraction
KCMO	Kansas City Medical Optimization
MRA	mineralocorticoid receptor antagonists

adjusting for confounding, predicting long-term outcomes, and benchmarking quality-improvement initiatives.^{8–11}

Given this proliferation of HFrEF GDMT scoring systems, we undertook a scoping review to catalogue and characterize all published HF GDMT scores and summarize their application in clinical studies.

METHODS

The scoping review was conducted according to the Joanna Briggs Institute Manual for Evidence Synthesis, and findings are reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews.^{12,13} All data and supporting materials have been provided with the published article. This study is a scoping review of published articles; as such, ethics review board approval was not required.

Eligibility Criteria

Concept, Context, and Design

Eligible articles reported at least 1 scoring system quantifying HF pharmacotherapy (whether or not it was labeled as a GDMT, optimal medical therapy, or OMT score) applied in a

clinical study. Specifically, we considered scores that quantified utilization of individual HF medication classes and dosing. We included observational or interventional clinical studies reported in full-text articles from any health care setting, including outpatient clinics, inpatient wards, and community-based practices. We excluded commentaries, editorials, and other opinion documents such as consensus recommendations, narrative reviews, editorials and commentaries that did not evaluate a GDMT within a patient cohort. Therefore, we excluded GDMT scores that were described or proposed, but not evaluated in any clinical study. We did not apply any language restriction.

Participants

We included manuscripts focusing on adults with HF across the spectrum of left ventricular ejection fraction. Though most published articles on GDMT scores focus on HFrEF, we also sought any GDMT scores designed for patients with HF with an ejection fraction >40%. We excluded studies limited to patients under consideration for mechanical circulatory support or heart transplantation and those focused exclusively on palliative, end-of-life, or advanced-care planning. Studies in which the patient's HF subtype was not clearly specified were also excluded.

Database Search and Study Selection

We searched MEDLINE, Embase, and CENTRAL from October 5, 2020 (publication date of the original HFC GDMT score) to March 1, 2025. The full search queries for each database are included in the Supplemental Methods. We also performed forward-citation searches using Web of Science of 3 key relevant publications: the 2020 HFC article describing the original GDMT score and 2 follow-up publications describing revisions to the HFC GDMT score.^{6,7,14}

Search results were imported into Covidence (Veritas Health Innovation, Melbourne, Australia), and duplicates were removed automatically. Two reviewers (A.D. and R.D.T.) independently screened titles and abstracts, then full texts, for eligibility. Disagreements were resolved by discussion.

Data Charting Process, Data Items, and Synthesis

Two reviewers (A.D. and R.D.T.) independently extracted data using a piloted, standardized form in Google Sheets. Discrepancies were resolved by consensus. Extracted variables included bibliographic details, study design, setting, participant characteristics, description of the GDMT score (components, weighting, scale, thresholds), purpose of score application, and reported associations with clinical outcomes (all-cause mortality, HF hospitalization, worsening HF events, and quality of life). Extracted data were tabulated and synthesized narratively.

RESULTS

From 544 records, we included 29 articles, including the 3 key HFC publications and 26 clinical studies (Figure 1).^{6,7,14}

Characteristics of Included Studies

The 26 included studies enrolled a total 354 281 participants (Table 1). The median sample size was 942 (range

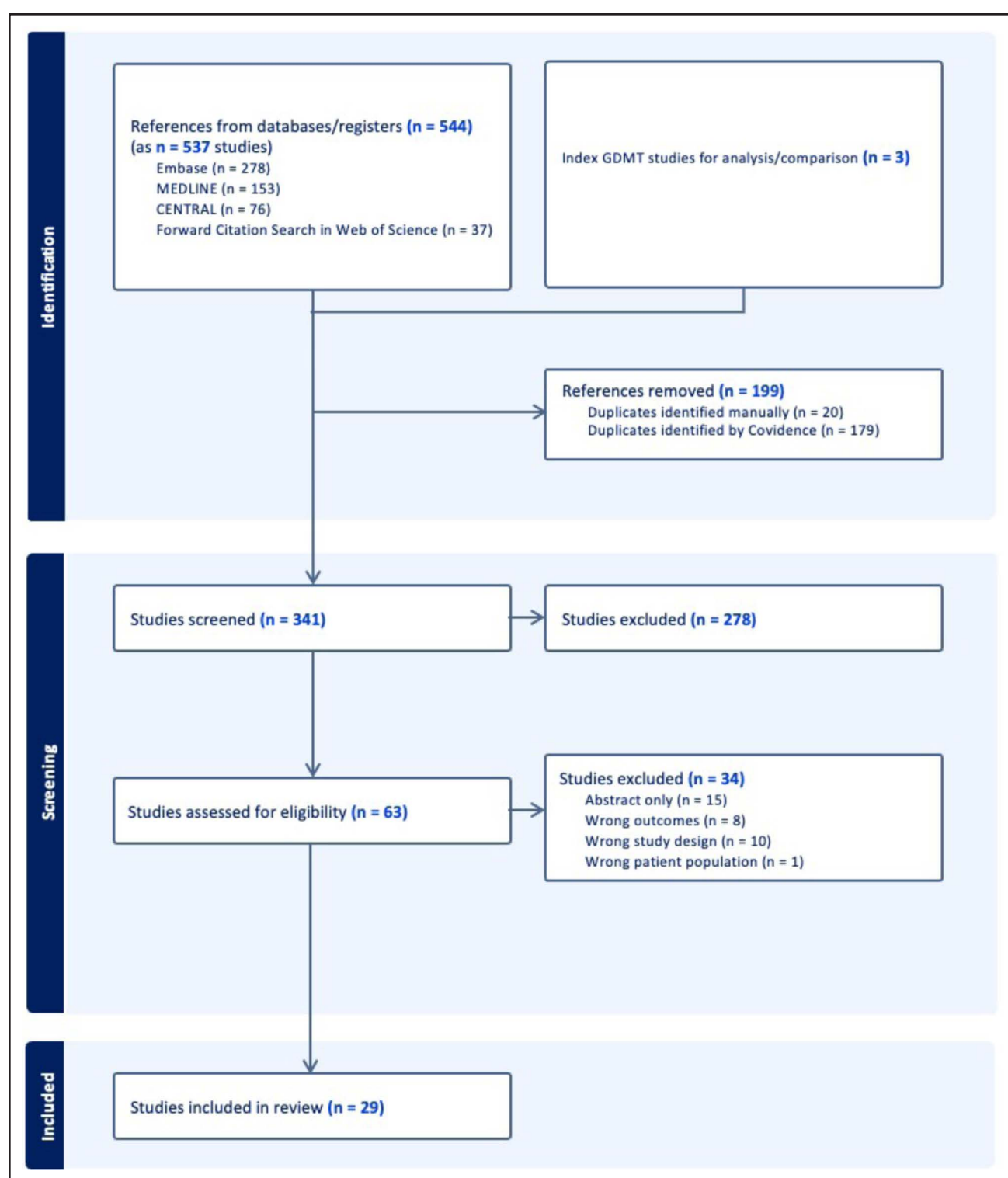


Figure 1. PRISMA flow diagram for study inclusion.

PRISMA study selection process. Database searches identified 544 records corresponding to 537 unique studies from Embase (n=278), MEDLINE (n=153), CENTRAL (Cochrane Central Register of Controlled Trials; n=76), and forward citation searching in Web of Science (n=37), along with 3 known landmark index guideline-directed medical therapy (GDMT) studies for analysis and comparison. After removing 199 duplicates, 341 studies were screened; 278 were excluded at the title/abstract. Sixty-three full texts were assessed, and 34 were excluded. Twenty-nine studies were included in the review.

Table 1. Characteristics of Included Studies

Study	n	HF Subtype	Setting
Observational studies			
Wahid et al ¹⁵	66 372	HFrEF	Outpatient
Johansen et al ¹¹	26 779	HFrEF	Outpatient
Martyn et al ¹⁶	6454	HFrEF	NR
Matsukawa et al ¹⁷	1054	HFrEF, HFmrEF	Inpatient
Rastogi et al ¹⁸	702	HFrEF, HFmrEF, HFpEF	Hospital discharge
Agrawal et al ¹⁹	829	HFrEF	Hospital discharge
Asano et al ²⁰	426	HFrEF, HFmrEF	Hospital discharge
Khan et al ⁹	4532	HFrEF	Outpatient
Kresoja et al ²¹	1072	HFrEF	Outpatient
Ladhar et al ²²	81	HFrEF	Outpatient
Matsukawa et al ²³	372	HFrEF, HFmrEF	Inpatient
Polsinelli et al ²⁴	41 161	HFrEF	Hospital discharge
Rao et al ²⁵	4582	HFrEF	Outpatient
Steverson et al ²⁶	108 752	HFrEF	Outpatient
Sueta et al ²⁷	177	Mixed	Inpatient
Sykora et al ²⁸	111	HFrEF, HFmrEF	Outpatient
Yan et al ²⁹	180	HFrEF	Outpatient
Jacobs et al ³⁰	82 637	HFrEF	Hospital discharge
Saveski et al ³¹	1319	HFrEF	Outpatient
Randomized controlled trials			
Butt et al ³²	4744	HFrEF	Outpatient
Bhatt et al ³³	198	HFrEF	Inpatient
Pabon et al ³⁴	1151	HFimpEF	Outpatient
Rao et al ³⁵	91	HFrEF	Inpatient
Romero et al ³⁶	55	HFrEF	Outpatient
Verma et al ³⁷	300	HFrEF	Outpatient
Man et al ³⁸	150	HFrEF	Outpatient

HF indicates heart failure; HFimpEF, heart failure with improved ejection fraction; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; and HFrEF, heart failure with reduced ejection fraction.

55–108,752). Of the included studies, 19 were observational cohorts, and 7 were randomized controlled trials. The study setting was outpatient in 15 studies; inpatient or at hospital discharge in 5, and unspecified in 1 study.¹⁶ Twenty-three studies were restricted to HFrEF, 6 included HF irrespective of left ventricular ejection fraction, and 1 specified inclusion of patients with HF with improved ejection fraction.

Characterizing Available GDMT Scores

Across the 26 studies, we identified 25 distinct HFrEF GDMT scores with prominent examples summarized in Figure 2 and a full list of scores available in Table S1. Below, we provide a synthesis of patterns and trends captured between these scores. We did not identify any GDMT scores designed for HF with mildly reduced or preserved ejection fraction.

HFC GDMT Score and Its Derivatives

The original HFC GDMT score, created before guideline recommendations for use of SGLT2Is in HFrEF, assigns up to 6 points based on use of an ACE inhibitor/ARB/ARNI, a beta-blocker, an MRA, and hydralazine-isosorbide dinitrate (Figure 2) and is intended for use in HFrEF populations.⁶ This score was revised in 2022, increasing the total score range to 0 to 12 by adding SGLT2Is, ivabradine, and vericiguat.⁷ Neither of these scores has been used unmodified in any of the included studies.

Thirteen variations of the HFC GDMT score (modified HFC GDMT scores v1–13), ranging in scores from 6 to 12 maximal attainable points (with 2 scores transformed to a value 0%–100%), were used across 16 studies.^{11,16,17,19,20,22–27,29,31,32,34,35} While retaining the additive logic of the original HFC GDMT score, they incorporated various changes to broaden therapeutic breadth or increase dose granularity.^{11,16,17,19,20,22–27,29,31,32,34,35} Modified HFC GDMT scores mainly differed in the weight of ARNIs and MRAs within the score, whether they included SGLT2Is or other agents (hydralazine-nitrate, ivabradine, vericiguat), and the incorporation of contraindications or documented intolerance (ie, maximum-tolerated GDMT). All but 1 score incorporating hydralazine-nitrate, ivabradine, and vericiguat assigned each agent a single point.

Other GDMT Scores

We identified 11 additional GDMT scores that did not cite the original HFC GDMT score manuscripts.^{9,15,18,21,26,28,30,33,36–38} All 11 scores captured ACE inhibitor/ARB, β -blocker, and MRA, whereas 8 also included ARNI and 4 also included SGLT2I. Scores varied in how they quantified uptake, dosing, and active optimization of GDMT. The presentation of scores ranged from changes from baseline (–8 to +8),³³ to percent of target dose achieved (0%–100%; eg, Kansas City Medical Optimization [KCMO] score).^{9,28}

Application of GDMT Scores in Clinical Studies

Included studies used GDMT scores to address various research objectives (Table 2). GDMT scores were most commonly used to describe baseline GDMT utilization (14 studies),^{9,11,15,16,19–21,25,27,30–32,34,35} followed by associations with covariate predictors (11 studies).^{9,16–20,23–25,30,32} Additionally, GDMT scores were used to define subgroups (2 studies),^{32,34} understanding hospital-level variation (2 studies),^{24,30} and describe medication utilization time-trends (2 studies).^{15,28} A GDMT score threshold was used as an eligibility criterion in 1 trial³⁷; the DASH-HF trial randomized patients with an Optimization Potential Score $\leq 5/10$ to a quality-improvement GDMT optimization intervention versus usual care.³⁷

GDMT score	ACEI/ARB	ARNI	BB	MRA	SGLT2i	OTHERS	Score range	Previous applications
HFC (2020)	1 ^{1-49%}	2	1 ^{1-49%}	1	-	1 ^{H-N}	0-6	-
	2 ^{≥50%}		2 ^{≥50%}					
KCMO (2024)	% target dose	-	% (TDD)/target	100%	-	-	0-100%	• Descriptor • Association with covariates
mHFC 2022 (Fiuzat)	1 ^{1-49%}	3	1 ^{1-49%}	2	2	- or 1 each ^{H-N, I, V}	0-9 or 0-12	• Descriptor • Association with covariates • Predictor of HF endpoints • Study endpoint • Medication utilization time trends
	2 ^{≥50%}		2 ^{≥50%}					
OPS (2023)	1 ^{sub-target}	3	1 ^{sub-target}	1 ^{sub-target}	2	-	0-10	• Study endpoint • Eligibility criterion
	2 ^{target}	4	2 ^{target}	2 ^{target}				
In-hospital GDMT optimization (2023)	-2 ^{Discontinue}	+2 ^{ARNI switch}	-2 ^{Discontinue}	-2 ^{Discontinue}	-2 ^{Discontinue}	-	-8 to 8	• Study endpoint
	-1 ^{Downtitrate}		-1 ^{Downtitrate}	-1 ^{Downtitrate}				
	+1 ^{Uptitrate}		+1 ^{Uptitrate}	+1 ^{Uptitrate}	+2 ^{Initiate}			
	+2 ^{Initiate}		+2 ^{Initiate}	+2 ^{Initiate}				
Modified C4P (2025)	2 to <3 ^{<50%}	4.5 to <5.5 ^{<50%}	2 to <3 ^{<50%}	2 to <3 ^{<50%}	3-4	-	0-18	• Predictor of HF endpoints
	3-4 ^{≥50%}	5.5-6 ^{≥50%}	3-4 ^{≥50%}	3-4 ^{≥50%}				
ΔGDMT (2024)	0-1 ^{%target dose}	1 ^{ARNI switch}	0-1 ^{%target dose}	0-1 ^{%target dose}	1	1 ^{iron deficiency screening or supplementation}	0-100%	• Study endpoint

Figure 2. Key guideline-directed medical therapy scores.

The figure includes studies from both Heart Failure Consortium (HFC) derivatives and original studies. Numbers in the medication columns indicate the value of each medication/dose in each scoring algorithm. Superscripts indicate the criteria within medication classes (eg, $\geq 50\%$ of target dose) for within-class scoring differences. The full list of included GDMT scores is in Table S1. ACE inhibitor indicates angiotensin-converting enzyme Inhibitor; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor-neprilysin inhibitor; BB, β -blocker; C4P, comprehensive cardiac care performance program; ΔGDMT, change in guideline-directed medical therapy; GDMT, guideline-directed medical therapy; H-N, hydralazine-nitrate; HF, heart failure; HFC, Heart Failure Consortium; I, ivabradine; KCMO, Kansas City Medical Optimizer; MRA, mineralocorticoid receptor antagonist; OPS, optimization performance score; SGLT2i, sodium-glucose cotransporter-2 inhibitor; and V, Vericiguat.

Association Between GDMT Scores and Clinical Outcomes

Thirteen studies evaluated the association between GDMT score and subsequent HF end points, including all-cause mortality, HF-related hospital readmissions, and composite outcomes (Table 3).^{11,17-20,23-28,32} Six studies assessed GDMT as a continuous predictor for each 1-point or 1-SD change in score. Stevenson et al²⁶ reported a significant $\approx 10\%$ reduction in the hazards for all-cause mortality (adjusted) and the composite of cardiovascular mortality or HF hospitalization (unadjusted)

per 1-SD improvement in each of the modified HFC GDMT score v2, continuous 4 pillar score, and modified continuous 4 pillar score. Similarly, Kresoja et al²¹ and Johansen et al¹¹ reported a 10% reduction in the hazards for all-cause mortality per 1-point improvement in their respective GDMT scores. The modified KCMO (range, 0%–100%) was associated with a 3% reduction in the hazards for their composite outcome per 1% change in GDMT score.²⁸

Seven studies dichotomized GDMT into higher- and lower-intensity categories, and 1 study assessed GDMT

Table 2. Application of GDMT Scoring Systems in Included Studies

Score	Application of the GDMT score and references							
	Descriptor (n=13)	Predictor of HF end points (n=11)	Association with covariates (n=10)	Study end point (n=8)	Medication utilization time trends (n=2)	Subgroup variable (n=2)	Hospital level variation (n=2)	Eligibility criterion (n=1)
Modified HFC v1	✓17,23,27,35	✓17,23,27,35	✓17,23,27,35	✓17,23,27,35	✓17,23,27,35			
Modified HFC v2		✓26						
Modified HFC v3	✓34					✓34		
Modified HFC v4	✓25	✓25	✓25					
Modified HFC v5		✓24	✓24				✓24	
Modified HFC v6	✓32	✓32	✓32			✓32		
Modified HFC v7	✓11	✓11						
Modified HFC v8	✓31							
Modified HFC v9	✓20	✓20	✓20					
Modified HFC v10				✓22				
Modified HFC v11	✓19	✓19	✓19					
Modified HFC v12	✓16		✓16					
Modified HFC v13				✓29				
Guideline Adherence		✓18	✓18					
Wahid 2022	✓15				✓15			
KCMO	✓9		✓9					
Romero 2023				✓36				
QTO	✓30		✓30				✓30	
OPS				✓37				✓37
Kresoja 2024	✓21							
In-Hospital GDMT Optimization				✓33				
Modified C4P		✓26						
Modified KCMO		✓		✓28				
ΔGDMT				✓38				

This table summarizes the applications of GDMT scoring systems across included studies, indicating whether each score was used as a descriptor, predictor of HF end points, associated with covariates, assessed as a study end point, or applied to evaluate medication utilization time trends, subgroup variables, hospital-level variation, or eligibility criteria. C4P indicates continuous 4 pillar; GDMT, guideline-directed medical therapy; HF, heart failure; HFC, heart failure collaboratory; KCMO, Kansas City Medical Optimization; OPS, optimization potential score; and QTO, quadruple therapy optimization.

score as both a continuous and categorical predictor.¹¹ Seven of 13 studies showed a statistically significant association between higher GDMT exposure and lower event rates^{11,17,18,21,24,26,28}; 1 detected no significant association²⁷; and the remaining 5 yielded mixed findings, with some end points reaching significance and others not.^{19,20,23,25,32}

GDMT Scores as Study End Points

Eight studies used a GDMT score as a primary or secondary end point in the evaluation of medication optimization interventions (Table 4).^{9,22,29,33,35–38} Interventions included virtual-team programs,^{33,35} pharmacist-delivered optimization,^{9,22} and remote or algorithm-based titration clinics.^{36–38} Two studies evaluating a virtual HF consultation service in inpatients demonstrated increased GDMT scores.^{33,35} Pharmacist-led initiatives were similarly effective across multiple GDMT scores.^{9,22} Telemedicine titration platforms showed mixed results with a digital-consult program, and a protocol-driven remote clinic

showing increases in GDMT scores, whereas the DASH-HF program did not significantly improve optimization potential scores compared with control in a randomized controlled trial.^{36–38} Only 2 of the 8 studies reported a target sample size based on a power calculation.^{37,38} Overall, these results demonstrated the responsiveness of various GDMT scores as process-of-care end points for use in trials of implementation/quality-improvement interventions.

DISCUSSION

In this scoping review, we identified 25 individual GDMT scores proposed since 2020, including 13 scores derived from the original HFC GDMT score. Although conceptually similar, these scores varied in how they weighed foundational HFrEF medication classes and doses, as well as whether they incorporated adjunctive medications such as hydralazine-nitrate, ivabradine, and vericiguat. These scores were used for various research

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Table 3. GDMT Score Association With Clinical Outcomes

Study	Comparison	End point assessed (dependent variables)	Effect size (95% CI)
Agrawal et al ¹⁹	Modified v11 HFC score ≥ 3 vs < 3	All-cause readmissions at 1 mo	Unadjusted 20.9% vs 28.0% ($P=0.02$)
		All-cause readmissions at 3 mo	Unadjusted 37.9% vs 42.3% ($P=0.2$)
Asano et al ²⁰	Modified v9 HFC score $\geq 5/7$ vs $< 5/7$	2-year composite end point (composite of cardiac mortality and HF rehospitalization)	Adjusted HR, 0.55 (0.32–0.94)
		All-cause mortality	Adjusted HR, 0.63 (0.33–1.16)
		HF rehospitalization	Adjusted HR, 0.57 (0.32–0.98)
Butt et al ³²	Modified v6 HFC tertile 2 (41–60%) vs 1 ($< 41\%$)	Worsening HF or CV mortality	Adjusted HR, 0.91 (0.78–1.07)
		HF hospitalization or CV mortality	Adjusted HR, 0.92 (0.78–1.08)
		All-cause mortality	Adjusted HR, 0.87 (0.71–1.05)
	Modified v6 HFC tertile 3 ($> 60\%$) vs 1	Worsening HF or CV mortality	Adjusted HR, 0.83 (0.71–0.97)
		HF hospitalization or CV mortality	Adjusted HR, 0.84 (0.71–0.98)
		All-cause mortality	Adjusted HR, 0.87 (0.72–1.05)
Steverson et al ²⁶	Modified v2 HFC per 1 SD change	All-cause mortality	Unadjusted HR, 0.82 (0.81–0.82) Adjusted HR, 0.89 (0.88–0.90)
		Composite of cardiovascular mortality or HF hospitalization	Unadjusted HR, 0.89 (0.88–0.90)
	C4P Score, per 1 SD change	All-cause mortality	Unadjusted HR, 0.85 (0.84–0.85) Adjusted HR, 0.91 (0.90–0.92)
		Composite of cardiovascular mortality or HF hospitalization	Unadjusted HR, 0.91 (0.90–0.92)
	Modified C4P Score, per 1 SD change	All-cause mortality	Unadjusted HR, 0.83 (0.83–0.84) Adjusted HR, 0.90 (0.89–0.91)
		Composite of cardiovascular mortality or HF hospitalization	Unadjusted HR, 0.90 (0.89–0.91)
		HF ACTION: Composite of all-cause mortality or hospitalization	HR, 0.61 (0.44–0.84)
Kresoja et al ²¹	Kresoja GDMT score, per 1 point increase	All-cause mortality	Unadjusted HR, 0.90 (0.86–0.95)
Johansen et al ¹¹	Modified v7 HFC 1-unit increase	All-cause mortality	Adjusted HR, 0.90 (0.88–0.92)
		Cardiovascular mortality	Adjusted HR, 0.87 (0.84–0.91)
	Modified v7 HFC score ≥ 4 vs 4	All-cause mortality	Adjusted HR, 0.72 (0.67–0.78)
		Cardiovascular mortality	Adjusted HR, 0.67 (0.57–0.78)
Matsukawa et al ²³	Increase GDMT (≥ 1 -point increase in modified v1 HFC score) vs no increase from admission to discharge	Composite of HF readmission or all-cause mortality	Adjusted HR, 0.41 (0.26–0.67)
		HF readmission	Adjusted HR, 0.45 (0.26–0.79)
		All-cause mortality	Adjusted HR, 0.52 (0.24–1.12)
Matsukawa et al ¹⁷	Modified v1 HFC Score ≥ 5 vs < 5	HF readmission	Adjusted HR, 0.476 (0.324–0.701)
		All-cause mortality	Adjusted HR, 0.241 (0.102–0.568)
		Composite of HF readmission or all-cause mortality	Adjusted HR, 0.431 (0.295–0.630)
Polsinelli et al ²⁴	Correlation between higher hospital-level GDMT score (modified HFC v5) and end point	Mortality	Adjusted Spearman ρ , -0.22 (-0.32 to -0.12)
		Mortality or HF rehospitalization	Adjusted Spearman ρ , -0.17 (-0.26 to -0.06)
Rao et al ²⁵	Modified v4 HFC ≥ 4 vs < 3	All-cause mortality	Adjusted HR, 0.77 (0.64–0.92)
		HF hospitalization	Adjusted HR, 1.01 (0.86–1.19)
		Composite of cardiovascular mortality or HF hospitalization	Adjusted HR, 0.94 (0.82–1.08)
	Modified v4 HFC 3 vs < 3	All-cause mortality	Adjusted HR, 0.94 (0.73–1.20)
		HF hospitalization	Adjusted HR, 1.00 (0.82–1.22)
		Composite of cardiovascular mortality or HF hospitalization	Adjusted HR, 0.94 (0.78–1.13)
Rastogi et al ¹⁸	Guideline adherence score ≥ 4 vs < 4	Composite of rehospitalization for HF or mortality (HFrEF subgroup)	Adjusted HR, 0.75 (0.57–0.97)
Sueta et al ²⁷	Modified v1 HFC ≥ 5 vs < 5	All-cause mortality	HR, 0.645 (0.413–1.006)
		HF hospitalization	Unadjusted HR, 0.790 (0.355–1.759)
Sykora et al ²⁸	Modified KCMO per 1-unit change	Composite of unplanned HF hospitalization, LVAD implantation, heart transplant, or death	HR, 0.97 (0.95–0.99)

This table summarizes the association between GDMT score categories and clinical outcomes across included studies. Comparisons reflect different cutoffs or tertiles of GDMT scores. End points assessed include all-cause readmissions, all-cause mortality, HF rehospitalization, and composite outcomes. Effect sizes are presented as unadjusted event rates or adjusted hazard ratios (HRs) with 95% CIs, as reported in the studies. C4P, continuous 4 pillar; GDMT, guideline-directed medical therapy; HF, heart failure; HFC, heart failure collaboratory; HR, hazard ratio; KCMO, Kansas City Medical Optimization; OPS, optimization potential score.

Table 4. GDMT Score as a Study End Point

Study	Comparison	End point	Effect size
Bhatt et al ³³	Virtual team guided care vs usual care	Rate of GDMT optimization during hospitalization (in-hospital GDMT optimization score)	Adjusted mean difference +1.2 (+0.7 to +1.8)
Ladhar et al ²²	Pharmacist-led medication optimization, from clinic admission to discharge	Modified HFC score v10, median (IQR)	Admission 6 (4–7) vs discharge 8 (7–8), <i>P</i> <0.001
		OPS, median (IQR)	Admission 5 (3–6) vs 8 (6–9), <i>P</i> <0.001
Sykora et al ²⁸	Pharmacist-led medication therapy management vs usual care	Modified KCMO, mean (SD)	74.5 (38.4) vs 51.6 (33.2), <i>P</i> <0.001
Man et al ³⁸	Remote digital consult vs usual care	ΔGDMT score, median at 12 wk	Intervention 1.19 vs 0.08; difference 0.75 (0.21–1.12)
		ΔGDMT Score over 12 wk based on usual care	Median, 0.08; IQR (0.00–1.00)
Rao et al ³⁵	Virtual team guided care vs usual care	Modified HFC score v1	Absolute mean difference +0.86 (0.03–1.69)
Romero et al ³⁶	Remote GDMT titration clinic vs usual care	Romero 2023 score, difference at 6 mo	+8.1% (+1.7% to 14.5%)
		Romero 2023 score, difference at 12 mo	+12.8% (+5.0% to +20.6%)
Verma et al ³⁷	DASH-HF clinic vs usual care	OPS, difference in change from baseline	Adjusted mean difference +0.3 (–0.1 to +0.7)
Yan et al ²⁹	Personalized GDMT optimization protocol	Modified HFC score v13, change from initial to final visit	4.7 vs 5.9 (<i>P</i> <0.001)

This table summarizes studies that used GDMT scores as a study end point, comparing different interventions (eg, pharmacist-led optimization, remote consult services, titration clinics, and personalized optimization protocols) against usual care. End points included changes in GDMT scores or specific score versions (eg, HFC, KCMO, OPS), with effect sizes expressed as adjusted mean differences, medians with IQRs, or absolute differences. GDMT indicates guideline-directed medical therapy; HFC, heart failure collaboratory; IQR, interquartile range; KCMO, Kansas City Medical Optimization; OPS, optimization potential score; and ΔGDMT, change in GDMT.

applications, including as eligibility criteria, stratification variables, predictors of subsequent HF-related morbidity and mortality, and more recently as clinical study end points.

Domains of Heterogeneity in GDMT Scoring and Current Pitfalls

The surge of GDMT scores proposed in the past 5 years highlights a need for a unified, standardized approach to quantify GDMT utilization in practice and in clinical trials. However, the substantial heterogeneity in existing scores illustrates a lack of consensus in the underlying theoretical and empirical justification for their development and use. The original HFC GDMT score was proposed as a standard measure for cross-trial comparisons of baseline “background” use of evidence-based pharmacotherapy. Conversely, the subsequent plurality of GDMT scores, including those modified from the HFC GDMT score and others, prevents cross-study comparison and complicates synthesis of evidence. The GDMT scores identified in this review have progressively expanded their scope to encompass SGLT2Is, ivabradine, vericiguat, and even correction of iron deficiency. However, the relative efficacy of these agents on HF-related morbidity and mortality differs, with most of these adjunctive agents providing no additional improvement in survival. Moreover, trials comparing different doses of HFrEF medications are generally limited, with evidence for ACEIs/ARBs showing a reduction in HF hospitalization—but not mortality—with higher versus lower doses, and limited evidence showing potential sex-based differences.³⁹ Practically, an additional point for dose titration of an ACE inhibitor/ARB would not yield the same survival advantage as switching

to ARNI or adding an MRA.^{40,41} Yet many GDMT scores treat these actions as interchangeable.⁹ Thus, 2 patients with identical GDMT scores may have markedly different prognoses depending on which therapies contribute to their score. To date, studies evaluating the association between GDMT scores and subsequent HF end points have not assessed heterogeneity based on individual score composition. Along similar lines, most GDMT scores do not consider whether a patient is eligible—and perhaps more importantly, whether they are ineligible due to contraindications—for each medication class. As a result, a patient may be assigned a lower score despite medications being contraindicated, which may falsely be attributed to poor performance, especially if used as a quality indicator. A recent study demonstrated the importance of careful adjudication of treatment eligibility when using quality and performance metrics, particularly when used to compare providers, specialties, and broader services.⁴²

Using Existing GDMT Scores in Future Studies: Fitting Form to Function

Given the abundance of recently proposed GDMT scores, we encourage researchers to use existing scores wherever possible, rather than developing their own. To that end, researchers can use the information from the present study to inform the selection of a GDMT score for their specific study design. For example, the optimization potential score may be suitable as both a simple eligibility criterion and study end point for an implementation trial of an intervention with an anticipated moderate to large effect size on utilization of HFrEF quadruple therapy. Conversely, researchers seeking a granular GDMT utilization end point with empirical evidence for lower HF

outcome risk with higher scores may select the modified KCMO score. Only 2 existing GDMT scores incorporated the concept of maximum-tolerated GDMT, which may be required for studies needing to account for absolute and relative contraindications to properly quantify treatment gaps based on patient-specific factors, particularly in the setting of older patients and those with multimorbidity.^{22,25} Future evaluations of existing GDMT scores may provide further empirical support to favor a specific score. Notably, these scores capture prescribed therapy, but not patient adherence or persistence. Studies using GDMT scores to predict HF outcomes may have their associations confounded by nonadherence. Similarly, implementation strategies that increase GDMT scores based on prescription records may not translate to improved clinical outcomes if they are not accompanied by adherence and longer-term persistence to prescribed therapies.

Further Refining and Validating GDMT Scores

Based on this review, any newly proposed or refinements to existing GDMT scores should have the strengths of existing scores like the 2022 modified HFC GDMT score or optimization potential score: it should only include medications with proven benefits on patient-centered outcomes, including death and HF hospitalization; account for dose-response, when known; weigh each medication and dose range proportional to its effect on these outcomes; and explicitly account for contraindications and intolerances. Ideally, scores should be calculable for core (eg, quadruple therapy) and expanded medications, as done for quadruple therapy in the 2022 modified HFC GDMT score.⁷ Given the abundance of existing scores, including several that include the above elements, new GDMT score proposals should be further validated and demonstrated to be superior to existing scores to justify adoption. Validation of new and existing GDMT scores should match their intended application: scores intended to predict HF events should be assessed for their association with subsequent death and HF hospitalizations (composite and individual components) in large representative data sets—for example, as done by Butt et al and Johansen et al¹¹—assessing discrimination, monotonicity and linearity, and potential threshold effects.³² Moreover, there is a need to validate GDMT scores according to established surrogate outcome validation frameworks to support their use as outcomes in implementation trials.⁴³

Future Directions

Our scoping review uncovered a proliferation of GDMT scoring systems, and future research should advance along 2 complementary paths. First, investigators may broaden the utility of existing scores by validating them in diverse HF populations and clinical contexts. The present review illustrates existing applications evaluated for the

included scores, which can inform further investigation. For example, inpatient registries reveal that initiation and continuation of GDMT during HF hospitalizations remain suboptimal.⁴⁴ GDMT scores may serve as quality improvement metrics to highlight and monitor HF medication optimization gaps across clinical settings. Furthermore, scoring systems must evolve in step with therapeutic standards that now extend beyond HFREF. The present study did not identify any GDMT scores designed for patients with mildly reduced or preserved ejection fraction. As evidence accrues for medication efficacy in these patients,^{45–47} analogous scoring systems may be useful along the left ventricular ejection fraction continuum.

Second, future GDMT scoring systems should be justified based on strong theoretical and empirical foundations and then rigorously validated. Thus, new scores should include only therapies that have demonstrated a benefit on end points that are meaningful to patients and health systems (with the specific outcome clearly defined). Furthermore, scores must evolve to reflect contemporary pharmacotherapeutic options and weigh individual treatment components relative to their efficacy. Moreover, any developed score must be practical and fit for purpose in its intended use. For example, a score intended for quality improvement should be easily extractable from electronic health records, while 1 intended as a trial end point should be sensitive to clinically meaningful changes. Future GDMT score implementations could leverage natural language processing methods to calculate scores and account for physiological intolerance and contraindications based on the electronic medical record.

Whether using an existing score or a new 1, authors should clearly define the scoring system and describe any modifications when it comes to publishing or presenting their results. They should justify changes and explain how medication dosages, thresholds, and contraindications are handled. This transparency would ensure reproducibility and allow clinicians and researchers to interpret and compare GDMT scores appropriately across studies.

Limitations

Our review has notable limitations. With the rapid proliferation of GDMT scores and inconsistent terminology, it is possible that our search missed unpublished scores embedded into electronic decision support tools, or those not clearly labeled as GDMT scores and not citing the original HFC score or other existing GDMT scores. Furthermore, the rapid evolution of both therapy and scoring means that recently published scores or conference abstracts may have been missed by our search cut-off.

Conclusions

This scoping review identified several GDMT scores developed to quantify HFREF GDMT optimization. Key

gaps in scores include heterogeneous weighting of drug classes and doses, minimal incorporation of intolerance to identify maximum-tolerated therapy, and variable incorporation of therapies beyond quadruple therapy. Future studies should focus on clear reporting of the selected GDMT score and the use of existing scores that incorporate relevant contemporary agents.

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Supplemental Material

Supplemental Methods
Table S1

REFERENCES

- McDonald M, Virani S, Chan M, Ducharme A, Ezekowitz JA, Giannetti N, Heckman GA, Howlett JG, Koshman SL, Lepage S, et al. CCS/CHFS heart failure guidelines update: defining a new pharmacologic standard of care for heart failure with reduced ejection fraction. *Can J Cardiol*. 2021;37:531–546. doi: 10.1016/j.cjca.2021.01.017
- Heidenreich PA, Bozkurt B, Aguilar D, Allen LA, Byun JJ, Colvin MM, Deswal A, Drazner MH, Dunlay SM, Evers LR, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: executive summary: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145:e876–e894. doi: 10.1161/CIR.0000000000001062
- Yancy CW, Jessup M, Bozkurt B, Butler J, Casey DE, Drazner MH, Fonarow GC, Geraci SA, Horwich T, Januzzi JL, et al. 2013 ACCF/AHA guideline for the management of heart failure: Executive summary: a report of the American College of Cardiology Foundation/American Heart Association task force on practice guidelines. *Circulation*. 2013;128:1810–1852. doi: 10.1161/CIR.0b013e31829e8807
- Ezekowitz JA, O'Meara E, McDonald MA, Abrams H, Chan M, Ducharme A, Giannetti N, Grzeslo A, Hamilton PG, Heckman GA, et al. 2017 comprehensive update of the Canadian Cardiovascular Society guidelines for the management of heart failure. *Can J Cardiol*. 2017;33:1342–1433. doi: 10.1016/j.cjca.2017.08.022
- Shahim B, Kapelios CJ, Savarese G, Lund LH. Global public health burden of heart failure: an updated review. *Card Fail Rev*. 2023;9:e11. doi: 10.15420/cfr.2023.05
- Abraham WT, Psotka MA, Fiuzat M, Filippatos G, Lindenfeld JA, Mehran R, Ambardekar AV, Carson PE, Jacob R, Januzzi JL, et al. Standardized definitions for evaluation of heart failure therapies: scientific expert panel from the Heart Failure Collaboratory and Academic Research Consortium. *JACC Heart Fail*. 2020;8:961–972. doi: 10.1016/j.jchf.2020.10.002
- Fiuzat M, Hamo CE, Butler J, Abraham WT, DeFilippis EM, Fonarow GC, Lindenfeld J, Mentz RJ, Psotka MA, Solomon SD, et al. Optimal background pharmacological therapy for heart failure patients in clinical trials. *J Am Coll Cardiol*. 2022;79:504–510. doi: 10.1016/j.jacc.2021.11.033
- Peri-Okonny PA, Mi X, Khariton Y, Patel KK, Thomas L, Fonarow GC, Sharma PP, Duffy CI, Albert NM, Butler J, et al. Target doses of heart failure medical therapy and blood pressure: insights from the CHAMP-HF Registry. *JACC Heart Fail*. 2019;7:350–358. doi: 10.1016/j.jchf.2018.11.011
- Khan MS, Chan PS, Sherrod CF, Ikemura N, Sauer AJ, Jones PG, Fonarow GC, Butler J, DeVore AD, Lund LH, et al. Generalizable approach to quantifying guideline-directed medical therapy. *Circ Heart Fail*. 2024;17:e011164. doi: 10.1161/CIRCHEARTFAILURE.123.011164
- Ahmad FS, Ahmad T, Allen LA. When the whole is the sum of its parts: validation of the Heart Failure Collaboratory medication composite score in Denmark. *Circ Heart Fail*. 2023;16:e010309. doi: 10.1161/CIRCHEARTFAILURE.122.010309
- Johannsen ND, Vaduganathan M, Zahir D, Fiuzat M, DeFilippis EM, Januzzi JL, Butler J, O'Connor CM, Abraham WT, Psotka MA, et al. A composite score summarizing use and dosing of evidence-based medical therapies in heart failure: a nationwide cohort study. *Circ Heart Fail*. 2023;16:e009729. doi: 10.1161/CIRCHEARTFAILURE.122.009729
- JB1 manual for evidence synthesis. 2024; doi: 10.46658/JBIMES-24-01
- Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, Moher D, Peters MDJ, Horsley T, Weeks L, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med*. 2018;169:467–473. doi: 10.7326/M18-0850
- DeFilippis EM, Fiuzat M. Putting the "optimal" in optimal medical therapy. *JACC Heart Fail*. 2021;9:39–41. doi: 10.1016/j.jchf.2020.08.016
- Wahid M, Aghanya V, Sepehrvand N, Dover DC, Kaul P, Ezekowitz J. Use of guideline-directed medical therapy in patients aged ≥ 65 years after the diagnosis of heart failure: a Canadian population-based study. *CJC Open*. 2022;4:1015–1023. doi: 10.1016/j.cjco.2022.08.003
- Martyn T, Saef J, Khot UN, Martinez KA, Brophy TJ, West L, Cristiani C, Block-Beach H, Hohman JA, Sobol T, et al. The utilization and impact of cardiovascular specialists on guideline-directed medical scores: an analysis of a diverse, multi-state, electronic health record-based registry. *Eur J Heart Fail*. 2023;25:2333–2336. doi: 10.1002/ehf2.14524
- Matsukawa R, Okahara A, Tokutome M, Itonaga J, Koga E, Hara A, Kisanuki H, Sada M, Okabe K, Kawai S, et al. A scoring evaluation for the practical introduction of guideline-directed medical therapy in heart failure patients. *ESC Heart Fail*. 2023;10:3352–3363. doi: 10.1002/ehf2.14524
- Rastogi T, Duarte K, Huttin O, Roubille F, Giered N. The prescription pattern of heart failure medications in reduced, mildly reduced, and preserved ejection fractions. *J Clin Med*. 2022;12:99. doi: 10.3390/jcm12010099
- Agrawal S, Alhaddad Z, Nabia S, Rehman OU, Kiyani M, Kumar A, Regmi N, Pingili A, Garg T, Allamaneni R, et al. Prescription patterns in management of heart failure and its association with readmissions: a retrospective analysis. *J Card Fail*. 2025;31:635–645. doi: 10.1016/j.cardfail.2024.08.059
- Asano T, Maeno Y, Nakano M, Taguri M, Miyasaka M, Nakai D, Miyazaki I, Nasu T, Tanimoto S, Masuda N, et al. Validation of a new scoring method to assess the efficacy of rapid initiation and titration of combination pharmacotherapy for patients hospitalized with acute decompensated heart failure with reduced and mildly reduced ejection fraction. *JCM*. 2024;13:2775. doi: 10.3390/jcm13102775
- Kresoja KP, Adamo M, Rommel KP, Stolz L, Karam N, Giannini C, Melica B, von Bardeleben RS, Butter C, Horn P, et al. Guideline-directed medical therapy assessment in heart failure patients undergoing percutaneous mitral valve repair. *ESC Heart Fail*. 2024;11:1802–1807. doi: 10.1002/ehf2.14705
- Ladhar S, Hawkins NM, Virani SA, Turgeon RD. Evaluation of PHARM-HF, a pharmacist-led heart failure medication titration clinic: A pre-post study. *Journal of the American College of Clinical Pharmacy*. 2024;7:416–425. doi: https://doi.org/10.1002/JAC5.1920
- Matsukawa R, Kabu K, Koga E, Hara A, Kisanuki H, Sada M, Okabe K, Okahara A, Tokutome M, Kawai S, et al. Optimizing guideline-directed medical therapy during hospitalization improves prognosis in patients with worsening heart failure requiring readmissions. *Circ J*. 2024;88:1416–1424. doi: 10.1253/circj.CJ-24-0265
- Polsinelli VB, Sun JL, Greene SJ, Chiswell K, Grunwald GK, Allen LA, Peterson P, Pandey A, Fonarow GC, Heidenreich P, et al. Hospital heart failure medical therapy score and associated clinical outcomes and costs. *JAMA Cardiol*. 2024;9:1029–1038. doi: 10.1001/jamacardio.2024.2969
- Rao VN, Hellkamp AS, Thomas LE, Fonarow GC, Fiuzat M, O'Connor CM, Spertus JA, Desai AS, Albert NM, Butler J, et al. Optimal medical therapy and outcomes among patients with chronic heart failure with reduced ejection fraction. *JACC Heart Fail*. 2024;12:1862–1875. doi: 10.1016/j.jchf.2024.05.026
- Steverson A, Calma J, Hsiao S, Sallam K, Varshney AS, Golbus JR, Heidenreich PA, Sandhu AT. Validating the association between composite metrics of guideline-directed medical therapy and clinical outcomes for patients with heart failure with reduced ejection fraction. *J Card Fail*. 2025;31:588–591. doi: 10.1016/j.cardfail.2024.08.054

27. Sueta D, Araki S, Usuku H, Fujisaki T, Kiyama T, Ishii M, Tabata N, Fujisue K, Kusaka H, Hanatani S, et al. Clinical outcomes of patients from older community hospitalized for heart failure in guideline-directed medical therapy era: Insights from the COMPASS-HF registry. *J Cardiol*. 2025;85:257–259. doi: 10.1016/j.jjcc.2024.09.006
28. Sykora D, Olson N, Churchill R, Kim BM, Bratcher M, Elwazir M, Young K, Ryan S, Brodin M, Anderson J, et al. Pharmacist medication titration program for patients with cardiac sarcoidosis and systolic heart failure: a retrospective cohort study. *J Am Heart Assoc*. 2024;13:e038965. doi: 10.1161/JAHA.124.038965
29. Yan CL, Snipelisky D, Velez M, Baran D, Estep JD, Bauerlein EJ, Thakkar Rivera N. Protocol-driven approach to guideline-directed medical therapy optimization for heart failure: a real-world application to recovery. *Am Heart J Plus*. 2024;45:100438. doi: 10.1016/j.ahjo.2024.100438
30. Jacobs JA, Ayodele I, Bress AP, Sterling MR, Pandey A, Derington CG, Zheutlin AR, Shah KS, Greene SJ, Alhanti B, et al. Social determinants of health and disparities in guideline-directed medical therapy optimization for heart failure. *Circ Heart Fail*. 2025;18:e012357. doi: 10.1161/CIRCHEARTFAILURE.124.012357
31. Saveski D, Kok M, Poon S, Rojas-Fernandez C, Virani SA, Honos G, McKelvie R. The Canadian Heart Failure (CAN-HF) Registry: a Canadian multicentre, retrospective study of outpatients with heart failure. *CJC Open*. 2025;7:1–9. doi: 10.1016/j.cjco.2024.09.014
32. Butt JH, Dewan P, DeFilippis EM, Biering-Sorensen T, Docherty KF, Jhund PS, Kosiborod MN, Martinez FA, Bengtsson O, Johansen ND, et al. Effects of dapagliflozin according to the heart failure collaborative medical therapy score: insights from DAPA-HF. *JACC Heart Fail*. 2022;10:543–555. doi: 10.1016/j.jchf.2022.03.009
33. Bhatt AS, Varshney AS, Moscone A, Claggett BL, Miao ZM, Chatur S, Lopes MS, Ostrominski JW, Pabon MA, Unlu O, et al. Virtual care team guided management of patients with heart failure during hospitalization. *J Am Coll Cardiol*. 2023;81:1680–1693. doi: 10.1016/j.jacc.2023.02.029
34. Pabon M, Claggett BL, Wang X, Miao ZM, Chatur S, Bhatt AS, Vaduganathan M, Fang JC, Desai AS, Jhund P, et al. Influence of background medical therapy on efficacy and safety of dapagliflozin in patients with heart failure with improved ejection fraction in the DELIVER trial. *Eur J Heart Fail*. 2023;25:1663–1670. doi: 10.1002/ehfj.3001
35. Rao VN, Shah A, McDermott J, Barnes SG, Murray EM, Kelsey MD, Greene SJ, Fudim M, DeVore AD, Patel CB, et al. In-hospital virtual peer-to-peer consultation to increase guideline-directed medical therapy for heart failure: a pilot randomized trial. *Circ Heart Failure*. 2023;16: doi: 10.1161/circheartfailure.122.010158
36. Romero E, Yala S, Sellers-Porter C, Lynch G, Mwathi V, Hellier Y, Goldman S, Rocha P, Fine JR, Liem D, et al. Remote monitoring titration clinic to implement guideline-directed therapy for heart failure patients with reduced ejection fraction: a pilot quality-improvement intervention. *Front Cardiovasc Med*. 2023;10:1202615. doi: 10.3389/fcvm.2023.1202615
37. Verma A, Fonarow GC, Hsu JJ, Jackevicius CA, Vaghaiwalla Mody F, Nguyen A, Amidi O, Goldberg S, Vetrivel R, Upparapalli D, et al. DASH-HF study: a pragmatic quality improvement randomized implementation trial for patients with heart failure with reduced ejection fraction. *Circ Heart Fail*. 2023;16:763–772. doi: 10.1161/circheartfailure.122.010278
38. Man JP, Koole MAC, Meregalli PG, Handoko ML, Stienen S, de Lange FJ, Winter MM, Schijven MP, Kok WEM, Kuipers DI, et al. Digital consults in heart failure care: a randomized controlled trial. *Nat Med*. 2024;30:2907–2913. doi: 10.1038/s41591-024-03238-6
39. Turgeon RD, Kolber MR, Loewen P, Ellis U, McCormack JP. Higher versus lower doses of ACE inhibitors, angiotensin-2 receptor blockers and beta-blockers in heart failure with reduced ejection fraction: systematic review and meta-analysis. *PLoS One*. 2019;14:e0212907. doi: 10.1371/journal.pone.0212907
40. She J, Lou B, Liu H, Zhou B, Jiang GT, Luo Y, Wu H, Wang C, Yuan Z. ARNI versus ACEI/ARB in reducing cardiovascular outcomes after myocardial infarction. *ESC Heart Fail*. 2021;8:4607–4616. doi: 10.1002/ehf2.13644
41. Tromp J, Ouwerkerk W, van Veldhuisen DJ, Hillege HL, Richards AM, van der Meer P, Anand IS, Lam CSP, Voors AA. A systematic review and network meta-analysis of pharmacological treatment of heart failure with reduced ejection fraction. *JACC Heart Fail*. 2022;10:73–84. doi: 10.1016/j.jchf.2021.09.004
42. Gencher JA, Salimian S, Turgeon RD, Tam P, Hawkins NM. Differences in indications, contraindications, and prescribing patterns for heart failure between specialties: the Vancouever Coastal Acute Heart Failure (VOCAL-AHF) registry. *Eur J Intern Med*. 2025;141:106486. doi: 10.1016/j.ejim.2025.106486
43. Manyara AM, Davies P, Stewart D, Weir CJ, Young AE, Blazeby J, Butcher NJ, Bujkiewicz S, Chan A-W, Dawoud D, et al. Reporting of surrogate endpoints in randomised controlled trial reports (CONSORT-Surrogate): extension checklist with explanation and elaboration. *BMJ*. 2024;386:e078524. doi: 10.1136/bmj-2023-078524
44. Zheng J, Sandhu AT, Bhatt AS, Collins SP, Pinn KM, Fonarow GC, et al. Inpatient use of guideline-directed medical therapy during heart failure hospitalizations among community-based health systems. *JACC Heart Fail*. 2024;13: doi: 10.1016/j.jchf.2024.08.004
45. Solomon SD, McMurray JJV, Vaduganathan M, Claggett B, Jhund PS, Desai AS, et al. Finerenone in heart failure with mildly reduced or preserved ejection fraction. *N Engl J Med*. 2024. doi: 10.1056/NEJMOA2407107/SUPPL_FILE/NEJMOA2407107_DATA-SHARING.PDF
46. Packer M, Zile MR, Kramer CM, Baum SJ, Litwin SE, Menon V, Ge J, Weerakkody GJ, Ou Y, Bunck MC, et al. Tirzepatide for heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2025;392:427–437. doi: 10.1056/nejmoa2410027
47. Kosiborod MN, Deanfield J, Pratley R, Borlaug BA, Butler J, Davies MJ, Emerson SS, Kahn SE, Kitzman DW, Lingvay I, et al; SELECT, FLOW, STEP-HFpEF, and STEP-HFpEF DM Trial Committees and Investigators. Semaglutide versus placebo in patients with heart failure and mildly reduced or preserved ejection fraction: a pooled analysis of the SELECT, FLOW, STEP-HFpEF, and STEP-HFpEF DM randomised trials. *Lancet*. 2024;404:949–961. doi: 10.1016/S0140-6736(24)01643-X